

Geometric Shapes: Hierarchy, Brain Processing, and 2D Geons

1. Graph vs. Hypergraph for Geometric Hierarchy

The hierarchical structure between geometric shapes is better thought of as a **directed acyclic graph (DAG)**, rather than a hypergraph.

Here is why the graph model is the most suitable and how it differs from a hypergraph in this context:

Why a Graph (DAG) is the Ideal Choice:

- In geometry, hierarchical relationships are based on the **inclusion of properties** (e.g., '*a square is a type of rectangle, which in turn is a type of parallelogram*'). This link is inherently **binary** (connecting a 'child' shape to a 'parent' shape).
- **Nodes:** Represent classes of geometric shapes (Square, Rectangle, Rhombus, Parallelogram).
- **Directed Edges:** Represent the 'is-a' relationship.
- **DAG Structure:** Since a square inherits properties from both a rectangle and a rhombus, the structure is not a simple tree, but a **Directed Acyclic Graph**. This allows multiple inheritance without creating cycles.

Why the Hypergraph is Less Suitable (But Has Another Use):

- A **hypergraph** generalizes a graph by allowing a 'hyperedge' to connect any number of nodes simultaneously, instead of just two.
- **For pure hierarchy:** The hypergraph introduces unnecessary complexity. Saying that 'Square, Rectangle, and Rhombus are linked' via a hyperedge loses the direction and specificity of the hierarchy (who includes whom).
- **When would a hypergraph make sense?** It would be useful if you wanted to map **transversal common properties** that do not follow a strict hierarchy. For instance, a hyperedge could group 'all shapes with perpendicular diagonals' (Rhombus, Square, Kite) or 'all cyclic shapes' (Rectangle, Square, Isosceles Trapezoid). In this case, you would be tracking geometric traits, not the hierarchical taxonomy.

Feature	Graph (DAG)	Hypergraph
Relationship Type	Binary and Directed ($A \rightarrow B$)	Multilateral (Set of nodes)
Hierarchical Representation	Excellent (Clearly shows subsumption)	Dispersive (Confuses classification with grouping)

Multiple Inheritance	Natively managed via multiple incoming edges	Complex to visualize asymmetrically
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2. Human Brain Hierarchy in Processing Geometric Shapes

Yes, such a hierarchy is well known. Cognitive neuroscience and neurophysiology have demonstrated that **the human brain does not process geometric shapes all at once, but through a strict sequential and parallel hierarchy.**

The Visual Hierarchy (Bottom-Up):

- **Low Level (V1 Area):** Neurons activate only for **oriented lines and edges** (horizontal, vertical, or oblique segments). A square here is just a fragmented set of 4 separate lines.
- **Mid Level (V2 and V3 Areas):** The brain joins the lines and detects **intersections and corners** (e.g., recognizing a 90-degree right angle).
- **High Level (V4 Area and Occipito-Temporal Complex):** Features are assembled to perceive the **global shape** (the closure or the boundary layout).

The Symbolic Hierarchy and the Regularity Effect (Top-Down):

Recent fMRI and MEG studies show that the human brain possesses a unique abstract geometric perception mode. When we perceive shapes, we activate a hierarchy based on **geometric regularity**. High-regularity shapes (squares, rectangles) are processed faster and with less cerebral effort than irregular shapes, because the brain translates them into an unconscious mathematical code in the **intraparietal sulcus (IPS)**.

Geon Theory and the 'Language of Thought':

- **Geon Theory (Biederman):** The brain breaks down any complex 3D object into a basic geometric alphabet of about 36 simple volumetric shapes called **geons**.
- **Language of Geometric Thought:** Humans describe shapes in their minds using a recursive mental program made of geometric primitives. The shorter the mental program (e.g., for a square), the higher the shape stands in the brain's recognition hierarchy.

3. Is There a Geon Theory for 2D Shapes?

Irving Biederman's original **Recognition-by-Components (RBC) Theory** was designed for 3D objects. However, **yes, a 2D equivalent exists** both within his theory and through alternative cognitive models.

2D Projections of Geons (Non-Accidental Properties):

To explain how the brain extracts 3D shapes from a flat 2D retinal image, Biederman identified an alphabet of **2D edge properties** that the brain looks for immediately because they are highly unlikely to occur by perspective accident:

- **Curvature:** Determining whether a line is straight or curved.
- **Parallelism:** Detecting if two lines run parallel.
- **Coterminal:** Finding where lines meet to form vertices (e.g., 'T', 'Y', 'L' junctions).
- **Symmetry:** Discerning if the 2D profile has mirror axes.
- **Collinearity:** Detecting if multiple segments follow the same straight path.

2D 'Geometric Primitives':

In cognitive psychology and computer vision, the 2D equivalent of geons is known as **geometric primitives**. Instead of combining volumes, the brain breaks down complex flat designs by combining standard basic shapes such as **segments, arcs, circles, rectangles, triangles, and ellipses**, analyzing their spatial relationships.

Structural Information Theory (SIT):

Introduced by Emanuel Leeuwenberg, SIT is the strongest alternative for the 2D domain. It states that the brain doesn't have a fixed catalog of shapes, but analyzes 2D figures by trying to **minimize the code needed to describe them** using three unconscious mathematical rules: *repetition*, *symmetry*, and *alternation*.